Critical Appraisal: A Radiomics Approach to Assess Tumour-Infiltrating CD8 Cells and Response to Anti-PD-1/PD-L1 Immunotherapy

Sun et al., The Lancet Oncology, 2018

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Background: Medical Problem & Importance

Problem: Predicting response to cancer immunotherapy

- Anti-PD-1 / anti-PD-L1 checkpoint inhibitors have transformed oncology (melanoma, lung, lymphoma. . .)
- **Only 20-50%** of patients with advanced solid tumours respond to treatment
- Need to identify *a priori* which patients will benefit

Known biology: Pre-existing tumour immune infiltration by **CD8+ cytotoxic T cells** correlates with response to PD-1/PD-L1 therapy

Three Immune Phenotypes

- **Immune-inflamed**: dense CD8 infiltrate, responds to immunotherapy
- **Immune-excluded**: T cells blocked at tumour periphery
- **Immune-desert**: low CD8 infiltrate, poor response

Background: State of the Art & Limitations

State of the Art at Time of Publication

- CD8 infiltration assessed by **immunohistochemistry (IHC)** or **RNA sequencing** from tumour biopsies
- **PD-L1 expression** used as a biomarker but two RCTs showed it was *not* reliably associated with response
- 4 published studies explored radiomics and immune infiltration, but:
  - No patients treated with immunotherapy
  - Only *one* had an external validation cohort
  - Evidence level remained low

Limitations of Existing Approaches

- Biopsies are **invasive**, spatially limited (sampling bias), and not repeatable
- Cannot capture **spatial heterogeneity** or longitudinal changes
- No validated imaging-based predictor of immunotherapy response existed

Objectives & Main Contributions

Objective

Develop and independently validate a **CT-based radiomic signature of tumour-infiltrating CD8 cells**, and evaluate its ability to predict tumour immune phenotype and clinical outcomes in patients treated with anti-PD-1/PD-L1 immunotherapy.

Main Contributions

1. **First** radiomics-based biomarker of CD8 tumour infiltration validated in *three* independent cohorts
2. Link demonstrated between CT texture features and **gene expression**, **pathology**, and **clinical response**
3. Radiomic score shown to be **independent prognostic factor** for overall survival under immunotherapy

Methodology: Segmentation & Feature Extraction

Two Volumes of Interest (VOIs):

1. **Tumour**: semi-automatically delineated by 3 radiation oncologists (ISOgray v4.1)
2. **Peripheral ring**: automated ± 2 mm dilation/shrinkage of tumour boundary (4 mm ring)

Motivation for ring: capture the tumour **microenvironment**, *where immune cells infiltrate from periphery*

Features extracted:

- 4 grey-level matrices (3D): GLRLM, etc.
- 39 features per VOI $\rightarrow$ **78 radiomic features**
- $+$ 5 location variables $+$ 1 technical variable (kVp)
- **Total: 84 input variables**

Why include kVp (peak kilovoltage)?

Textural features are **highly dependent** on image acquisition parameters. Including kVp corrects for inter-scanner variability in heterogeneous cohorts.

Why include VOI location?

Tissue origin has a major effect on tumour immune contexture: liver vs. adenopathy vs. lung lesions have different baseline textures.

Methodology: Response Variable & Machine Learning

Response Variable: CD8 Gene Expression Signature

- **MOSCATO**: RNA-seq quantified with Salmon (**TPM**); **TCGA**: quantified as **RPKM** (different methods, corrected only by mean/variance rescaling
- CD8 abundance estimated via **CD8B gene signature** (Becht et al., MCPcounter)); $\log_2$-transformed, binarised at the **median** for AUC computation

CD8B gene expression is a **computational surrogate**, not a direct measurement of CD8 cell infiltration. The full inference chain is: *imaging* $\rightarrow$ *gene expression* $\rightarrow$ *assumed CD8 count* $\rightarrow$ *clinical response*. Each link is an untested assumption.

Elastic-Net Regularised Regression

$$\hat{y} = a_1 X_1 + \cdots + a_i X_i + b$$

- L1 (lasso) + L2 (ridge) penalties: simultaneous **feature selection** and **coefficient shrinkage**
- $\alpha = 0.5$ (grid search), λ by cross-validation; **8 variables retained** from 84

Methodology: The 8-Variable Radiomic Signature

The elastic-net retained:

- **5 radiomic features**:
  - 1 first-order feature
  - 4 second-order features from the **Grey-Level Run Length Matrix (GLRLM)**
- **2 VOI location** variables: adenopathy, head & neck
- **1 technical variable**: peak kilovoltage

Intuitive interpretation

Relatively **homogeneous, hypodense** *tumours and peripheral rings* were associated with a **high CD8 score**, consistent with inflammatory infiltrate.
Heterogeneity/high grey levels may reflect necrosis or chaotic vascularisation

GLRLM: Why?

The **Grey-Level Run Length Matrix** quantifies texture *homogeneity/heterogeneity* i.e., how far uniform intensity values extend in a given direction. It captures the spatial structure of immune infiltrate patterns.

Critical Note

Only **8 of 84** variables retained: strong regularisation avoids overfitting but may also discard relevant features.

Validation Strategy

Four-Cohort Design

Cohort	Endpoint	Rationale
MOSCATO ($n = 135$)	AUC (**training**)	Build radiomic signature vs. CD8B RNA-seq
TCGA/TCIA ($n = 119$)	AUC (primary)	External genomic concordance in 5 cancer types
Immune phenotype ($n = 100$)	AUC	Inflamed vs. desert tumours (extreme phenotypes)
Immunotherapy cohort ($n = 137$)	ORR, OS, PFS	Clinical predictive/prognostic value

Strengths

- 3 truly *independent* val. cohorts
- Statistically different cohorts $\Rightarrow$ no overfitting to a tumour subset
- Pathologist masked to radiomic results

Limitations

- All cohorts retrospective
- TCGA: strict QC $\Rightarrow$ only 119/435 eligible
- Immune phenotype: labels defined by **tumour type** (e.g., melanoma = inflamed), not pathological confirmation $\rightarrow$ *circular* with the downstream clinical claim

Key Results: Radiomic Signature Performance

AUC Results (High vs. Low CD8)

- **MOSCATO training**: AUC $= 0.74$ (0.66–0.82), $p < 0.0001$
- **TCGA validation**: AUC $= 0.67$ (0.57–0.77), $p = 0.0019$
- **Immune phenotype**: AUC $= 0.76$ (0.66–0.86), $p < 0.0001$

Pathology Concordance (TCGA)

Spearman $\rho = 0.559$, $p = 0.00022$ (4 cancer types **pooled**: BLCA, LUAD, LUSC, LIHC; $n = 39$)
Not significant for HNSC ($p = 0.18$)

Critical Note on AUC

AUC of 0.67 in the TCGA validation set is **modest**: suggests limited discriminative power in a truly external, multi-centre, multi-cancer dataset. The authors acknowledge this is exploratory.

Sensitivity Analysis

Removing non-radiomic variables (kVp, VOI location) caused **worse** TCGA performance, confirming their necessity for heterogeneous cohorts.

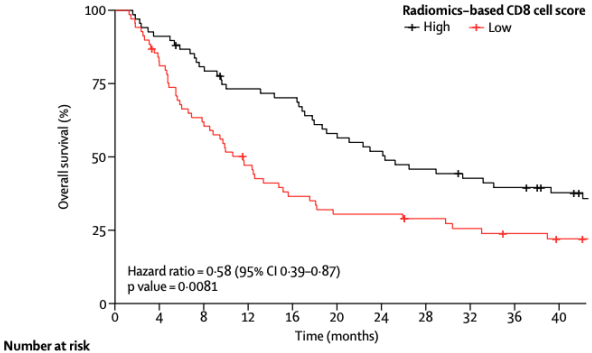
Key Results: Clinical Outcomes Under Immunotherapy

Objective Response Rate:

- At **3 months**: high score → higher ORR ($p = 0.049$)
- At **6 months**: higher ORR ($p = 0.025$) and disease control ($p = 0.013$)

Overall Survival ($n = 137$):

- Median OS: **24.3 vs 11.5 months**
- Univariate HR = 0.58 (0.39–0.87), $p = 0.0081$
- **Multivariate** HR = 0.52 (0.35–0.79), $p = 0.0022$



Number at risk (number censored)											
High	68 (0)	62 (1)	54 (1)	48 (2)	46 (2)	37 (2)	34 (2)	30 (4)	27 (4)	25 (5)	21 (6)
Low	69 (0)	55 (0)	41 (1)	31 (2)	24 (2)	20 (2)	20 (2)	17 (2)	15 (3)	13 (3)	11 (6)

Figure: Overall survival by radiomic score (high vs. low), Sun et al. 2018

- Median threshold: post-hoc, risks optimistic bias
- All liver lesions → **low score group** (confounder)
- 5 heterogeneous phase 1 trials

Conclusion

Strengths

- Multi-cohort with 3 truly independent validation sets
- Biologically motivated peripheral ring VOI
- Elastic-net: principled feature selection
- Sensitivity analysis confirming kVp/location necessity

Limitations

- Surrogate outcome (gene expression $\neq$ CD8 count); median binarisation
- No **calibration** reported $\rightarrow$ discrimination (AUC) only
- RPKM (TCGA) $\neq$ TPM (training) $\rightarrow$ quantification mismatch
- Retrospective; median threshold post-hoc
- Immune phenotype labels circular; liver lesion confounder
- No prospective validation

Thank you for your attention!

Appendix: Elastic-Net

$$\min_{\beta_0,\beta} \frac{1}{N}\sum_{i=1}^{N} l\Big(y_i, \beta_0 + \beta^T x_i\Big) + \lambda\left[(1-\alpha)\|\beta\|_2^2/2 + \alpha\|\beta\|_1\right]$$

The Ridge penalty shrinks the coefficients of correlated predictors towards each other while the Lasso tends to pick one of them and discard the others. The elastic net penalty mixes these two: if predictors are correlated in groups, an $\alpha = 0.5$ tends to either select or leave out the entire group of features.

Appendix: Radiomics Workflow

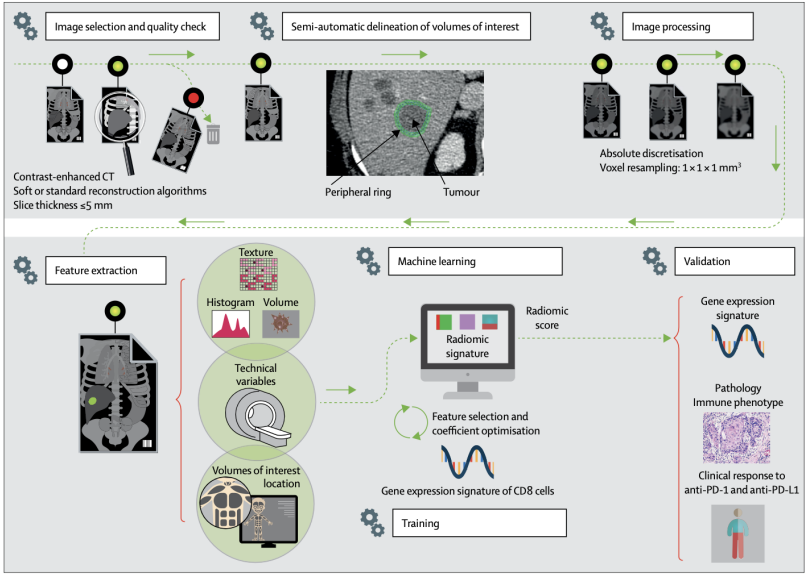


Figure: Radiomics Workflow